8. If $n((A \times B) \cap (A \times C)) = 8$ and $n(B \cap C) = 2$, then $n(A)$ is
   (1) 6 (2) 4 (3) 8 (4) 16

9. If $n(A) = 2$ and $n(B \cup C) = 3$, then $n[(A \times B) \cup (A \times C)]$ is
   (1) $2^3$ (2) $3^2$ (3) 6 (4) 5

10. If two sets $A$ and $B$ have 17 elements in common, then the number of elements common to the set $A \times B$ and $B \times A$ is
    (1) $2^{17}$ (2) $17^2$ (3) 34 (4) insufficient data

11. For non-empty sets $A$ and $B$, if $A \subset B$ then $(A \times B) \cap (B \times A)$ is equal to
    (1) $A \cap B$ (2) $A \times A$ (3) $B \times B$ (4) none of these.

12. The number of relations on a set containing 3 elements is
    (1) 9 (2) 81 (3) 512 (4) 1024

13. Let $R$ be the universal relation on a set $X$ with more than one element. Then $R$ is
    (1) not reflexive (2) not symmetric (3) transitive (4) none of the above

14. Let $X = \{1, 2, 3, 4\}$ and $R = \{(1, 1), (1, 2), (1, 3), (2, 2), (3, 3), (2, 1), (3, 1), (1, 4), (4, 1)\}$. Then $R$ is
    (1) reflexive (2) symmetric (3) transitive (4) equivalence

15. The range of the function $\frac{1}{1-2\sin x}$ is
    (1) $(-\infty, -1) \cup (\frac{1}{3}, \infty)$ (2) $(-1, \frac{1}{3})$ (3) $[-1, \frac{1}{3}]$ (4) $(-\infty, -1] \cup [\frac{1}{3}, \infty)$.

16. The range of the function $f(x) = |\lfloor x \rfloor - x|, x \in \mathbb{R}$ is
    (1) $[0, 1]$ (2) $[0, \infty)$ (3) $[0, 1)$ (4) $(0, 1)$

17. The rule $f(x) = x^2$ is a bijection if the domain and the co-domain are given by
    (1) $\mathbb{R}, \mathbb{R}$ (2) $\mathbb{R}, (0, \infty)$ (3) $(0, \infty), \mathbb{R}$ (4) $[0, \infty), [0, \infty)$

18. The number of constant functions from a set containing $m$ elements to a set containing $n$ elements is
    (1) $mn$ (2) $m$ (3) $n$ (4) $m + n$

19. The function $f : [0, 2\pi] \to [-1, 1]$ defined by $f(x) = \sin x$ is
    (1) one-to-one (2) onto (3) bijection (4) cannot be defined

20. If the function $f : [-3, 3] \to S$ defined by $f(x) = x^2$ is onto, then $S$ is
    (1) $[-9, 9]$ (2) $\mathbb{R}$ (3) $[-3, 3]$ (4) $[0, 9]$

21. Let $X = \{1, 2, 3, 4\}, Y = \{a, b, c, d\}$ and $f = \{(1, a), (4, b), (2, c), (3, d), (2, d)\}$. Then $f$ is
    (1) an one-to-one function (2) an onto function
    (3) a function which is not one-to-one (4) not a function